A directional prime mean bound in anchored gamma energy

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Abstract

We study the difference between two normalized prime-power means associated with the positive theta kernel for the Riemann xi function. On the minimum even homogeneous gamma-energy completion, represented by functions with zero theta-weighted mean on a fixed interval, we prove that its complete quadratic form is at most 3/25 of the gamma energy. The proof uses a direct anchor, five translation and reflection estimates, and double-exponentially small output and label tails. The finite numerator uses labels through 64, while both denominators retain every prime power. Disjoint output regions improve the core operator bound to 343/1000. Elementary rational comparisons suffice for the stated constants; the accompanying scripts check those comparisons, not the analytic proof. The result controls only the directional mean term. It gives no sign for the remaining parts of a full signed operator and does not prove the Riemann hypothesis.

Keywords. Theta kernel, prime-power means, nonlocal energy, weighted inequalities, exact rational bounds.

1 The form and the result

Let Λ be the von Mangoldt function, so that $\Lambda(p^k) = \log p$ for a prime p and an integer $k \geq 1$, and $\Lambda(n) = 0$ otherwise. Put $a_n = \Lambda(n)/\sqrt{n}$. For $x \geq 0$ define

$$t_j = \pi j^2 e^{2x}, \qquad \kappa(x) = e^{x/2} \sum_{j \geq 1} (4t_j^2 - 6t_j) e^{-t_j}, \qquad \kappa(-x) = \kappa(x). \tag{1.1}$$

The theta transformation gives its smooth even continuation. Each displayed summand on the nonnegative half-line is positive. The classical Fourier representation has the normalization

$$\int_{\mathbb{R}} \kappa(x) e^{izx} \, \mathrm{d}x = \xi(1/2 + iz), \qquad \xi(s) = \tfrac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s). \tag{1.2}$$

In particular, $\int_{\mathbb{R}} 2\cosh(x/2)\kappa(x) \, \mathrm{d}x = 1$. The kernel convention is compatible with the classical theta representation discussed, for example, in Romik [2]. The source identities needed below are stated with their derivations in Section 3.

Define

$$C(x) = \cosh(x/2), \quad W(x) = \frac{\kappa(x)}{2C(x)}, \quad \rho(s) = \frac{e^{-s/2}}{1 - e^{-2s}} \quad (s > 0), \tag{1.3}$$

$$P_- v(y) = \sum_{n \geq 2} a_n v(y - \log n), \quad P_+ v(y) = \sum_{n \geq 2} a_n v(y + \log n), \quad P = P_- + P_+, \tag{1.4}$$

$$L = P_- \kappa, \qquad R = P_+ \kappa. \tag{1.5}$$

The rows in (1.5) are always the complete positive rows. All prime powers, including powers of the same prime, retain their von Mangoldt weights. No denominator in this paper is truncated.

For a real even smooth compactly supported function f, set

$$\mathcal{E}_\Gamma(f) = \int_0^\infty \rho(s) \int_{\mathbb{R}} \kappa(x)\kappa(x+s)|f(x+s) - f(x)|^2 \, \mathrm{d}x \, \mathrm{d}s. \tag{1.6}$$

This is the integral over ordered edges $x < z$, or one half of the full symmetric double integral. Its normalization matters for the constant in the theorem. Write

$$a = \tfrac{1}{2}\log 2, \quad I = [-a, a], \quad K_I = \int_I \kappa, \quad \bar{f}_I = K_I^{-1} \int_I \kappa f, \quad g = f - \bar{f}_I. \tag{1.7}$$

Let $H_\mathcal{E}$ be the completion of the real even $C_c^\infty(\mathbb{R})$ core for the norm $\mathcal{E}_\Gamma^{1/2}$, with representatives anchored by $\int_I \kappa g = 0$. Section 4 constructs this representation; we do not identify this minimum completion with a larger maximal finite-energy domain.

On the core the normalized means and their difference are

$$\mu_L(f) = \frac{P_-(\kappa f)}{L}, \qquad \mu_R(f) = \frac{P_+(\kappa f)}{R},$$
$$Tf = \sqrt{WLR}\,(\mu_L(f) - \mu_R(f)), \qquad M(f) = \|Tf\|^2_{L^2(\mathbb{R})}. \tag{1.8}$$

The full difference annihilates constants, so $Tf = Tg$. Each mean by itself does not annihilate constants.

Theorem 1.1. *The map in* (1.8) *extends continuously from the even core to $H_\mathcal{E}$. Its full prime-power first moments converge absolutely at almost every output point for each anchored representative. On this space,*

$$M(f) \leq \frac{3}{25}\mathcal{E}_\Gamma(f). \tag{1.9}$$

In fact the proof gives

$$M(f) \leq \left[\left(\frac{343}{1000} + 10^{-48}\right)^2 + 9 \cdot 10^{-18}\right] \mathcal{E}_\Gamma(f) < \frac{3}{25}\mathcal{E}_\Gamma(f) \tag{1.10}$$

when $\mathcal{E}_\Gamma(f) > 0$. If the energy is zero, both forms vanish.

The estimate is an absorption bound for one positive quadratic form. It does not control a separate ratio-weighted variance term, a signed gamma compensation term, or a full signed spectral operator. In particular, (1.9) is not a proof of the Riemann hypothesis and makes no eigenstate exclusion claim. The relationship between positivity questions and explicit formulas is classical; see, for example, Connes and Consani [1]. No positivity theorem for the full explicit formula is assumed here. No conditional spectral gap or saved numerical spectral certificate enters the proof.

2 Elementary complete source estimates

The scalar comparisons used throughout are

$$\frac{25}{8} < \pi < \frac{22}{7}, \quad 20 < e^3 < 21, \quad \frac{1}{2} < \log 2 < \frac{7}{10}, \quad e > \frac{8}{3}, \quad e^{1/2} > \frac{79}{48}, \quad e^{3/4} > \frac{269}{128}. \tag{2.1}$$

Appendix A gives rational certificates for these elementary inequalities.

Lemma 2.1. *Let κ_1 be the $j = 1$ term in (1.1). For $r \geq 0$,*

$$\kappa_1(r) \leq \kappa(r) < \frac{251}{250}\kappa_1(r), \qquad \kappa(r) \leq K e^{9r/2 - \pi e^{2r}}, \qquad K = \frac{4\pi^2}{1 - 16e^{-3\pi}} < 40. \tag{2.2}$$

Moreover, $\kappa(r) < 1$ for every real r.

Proof. Dividing every positive term after the first by the first term and comparing successive terms for $j \geq 2$ bounds the relative tail by

$$\frac{16e^{-3\pi}}{(1 - 3/(2\pi))(1 - (3/2)^4 e^{-5\pi})} < \frac{16/8000}{(13/25)(1 - 81/(16 \cdot 20^5))} = \frac{2560000}{665598947} < \frac{1}{250}.$$

To obtain the second bound, discard the negative $-6t_j$ term and sum $j^4 e^{-\pi(j^2-1)}$. Its successive ratio is at most $16e^{-3\pi}$. Thus this bound includes the full theta series, and $K < 4(22/7)^2(500/499) = 968000/24451 < 40$.

Put $t = \pi e^{2r}$. Then

$$\kappa_1(r) = 4(t/\pi)^{1/4} t(t - 3/2)e^{-t}.$$

Its logarithmic derivative with respect to t is $5/(4t) + 1/(t - 3/2) - 1$, which is negative for $t \geq 16/5$. On $\pi \leq t \leq 16/5$, use $(t/\pi)^{1/4} < (101/100)$, $t - 3/2 \leq 17/10$, and $e^t > e^{25/8} > 20(9/8) = 45/2$. The same bound therefore controls the maximum over all $t \geq \pi$:

$$\kappa(r) < \frac{251}{250} 4 \frac{101}{100} \frac{16}{5} \frac{17}{10} \frac{2}{45} = \frac{3447736}{3515625} < 1.$$

Evenness gives the assertion on the negative half-line. □

Lemma 2.2. *For $y \geq 0$ and $n \geq 3$,*

$$\frac{a_n \kappa(y + \log n)}{a_2 \kappa(y + \log 2)} \leq \frac{8}{7} \frac{\Lambda(n)}{\log 2} \left(\frac{n}{2}\right)^4 e^{-\pi e^{2y}(n^2-4)}. \tag{2.3}$$

The full row R is decreasing on $[0, \infty)$, and

$$R(y) < \frac{1001}{1000} a_2 \kappa(y + \log 2), \qquad R(0) < \frac{3}{2000}, \qquad R(a) < \frac{1}{30000000}. \tag{2.4}$$

Proof. Use the complete envelope in the numerator and retain $n = 2, j = 1$ in the denominator. The resulting constant is bounded by

$$\frac{1}{(1 - 16e^{-3\pi})(1 - 3/(8\pi))} < \frac{500}{499} \frac{25}{22} < \frac{8}{7}.$$

Since $\Lambda(n) \leq \log n \leq n/2$ for $n \geq 2$, one has $\Lambda(n)/\log 2 < n$. The successive ratio of $n^5 e^{-\pi(n^2-4)}$ for $n \geq 3$ is less than $(4/3)^5/20^7 < 1/2$. Consequently the sum of the right side of (2.3) is less than

$$\frac{1}{14} \frac{2 \cdot 3^5}{20^5} = \frac{243}{22400000} < \frac{1}{1000}.$$

Every theta summand in $R(y)$ has $t \geq 4\pi$ and logarithmic derivative with respect to its source argument $9/2+3/(t-3/2)-2t < 0$. Positive partial sums, followed by their limit, prove monotonicity without exchanging an infinite derivative.

For the endpoints use $140/99 < \sqrt{2} < 99/70$, so that $a_2 < 99/200$, together with $4\pi > 25/2$. The first estimate is

$$R(0) < \frac{1001}{1000}\frac{99}{200}\frac{251}{250}\frac{99}{70}\frac{64(22/7)^2 - 24(25/8)}{20^4(79/48)} = \frac{28812434658579}{19355000000000000} < \frac{3}{2000}.$$

For $R(a)$ use $a_2 < 1/2$, $2^{3/4} < 27/16$, $2^{27/4} < 108$, and $8\pi > 25$. The complete envelope gives

$$R(a) < \frac{1001}{1000}\frac{2160}{20^8(8/3)} = \frac{81081}{2560000000000} < \frac{1}{30000000}.$$

□

3 Unconditional source identities and complete row estimates

This section proves the entry estimates used below. All prime sums contain all prime powers. No conjectural assertion about the critical-line location or simplicity of the nontrivial zeros, or about the sign of an explicit-formula quadratic form, is used.

3.1 Normalization, the pole, and the ground row

Let

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \Xi(z) = \xi(\tfrac{1}{2}+iz), \qquad \widehat{u}(z) = \int_{\mathbb{R}} u(x)e^{-izx}\,\mathrm{d}x.$$

Jacobi inversion gives the smooth even continuation of the displayed positive theta series for κ. For completeness, if $\vartheta(t) = \sum_{j\geq 1} e^{-\pi j^2 t}$, that series, with $t = e^{2x}$, is

$$\kappa(x) = e^{x/2}\{4t^2\vartheta''(t) + 6t\vartheta'(t)\}.$$

Jacobi inversion cancels the leading $t^{-1/2}$ term in the braces at zero. Mellin substitution and two integrations by parts, first for $\Re s > 1$, therefore give

$$\int_{\mathbb{R}} \kappa(x)e^{(s-1/2)x}\,\mathrm{d}x = \frac{s(s-1)}{2}\int_0^\infty t^{s/2-1}\vartheta(t)\,\mathrm{d}t = \xi(s).$$

Both sides continue to entire functions. In particular,

$$\widehat{\kappa} = \Xi, \qquad \int_{\mathbb{R}} 2C(x)\kappa(x)\,\mathrm{d}x = 1. \tag{3.1}$$

These are classical theta–Mellin identities; see [2].

Write

$$D_\Gamma v(x) = \int_0^\infty \rho(s)\{2v(x) - v(x+s) - v(x-s)\}\,\mathrm{d}s, \qquad c_\Gamma = \log\pi + \gamma + \frac{\pi}{2} + 3\log 2.$$

The difference inside the integral is kept coupled at zero. The complete source row is

$$P\kappa = C + (D_\Gamma - c_\Gamma)\kappa, \qquad P = P_- + P_+. \tag{3.2}$$

We record explicitly its unconditional origin, including the pole. For real even compact smooth physical tests put

$$\mathcal{Q}(u) = \int_0^\infty \rho(s)\|u(\cdot+s)-u\|_{L^2(\mathbb{R})}^2 \,\mathrm{d}s - c_\Gamma\|u\|_{L^2(\mathbb{R})}^2 - 2\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\langle u, u(\cdot+\log n)\rangle + 2\left|\int_{\mathbb{R}} Cu\right|^2.$$

The polarized classical explicit formula, in this normalization, is

$$\mathcal{Q}(u,v) = \sum_{\rho_\zeta} \widehat{u}(\overline{z_{\rho_\zeta}})\widehat{v}(z_{\rho_\zeta}), \qquad z_{\rho_\zeta} = (\rho_\zeta - \tfrac{1}{2})/i, \tag{3.3}$$

with all nontrivial zeros and their multiplicities. The archimedean normalization and the two pole terms are those of [1, Appendix B]. More explicitly, the archimedean expression before rewriting is

$$-\{\log(4\pi)+\gamma\}\|u\|_2^2 - 2\int_0^\infty \frac{e^{s/2}\langle u, u(\cdot+s)\rangle - \|u\|_2^2}{e^s - e^{-s}}\,\mathrm{d}s.$$

The identity $2\int_0^\infty (e^{s/2}-1)/(e^s-e^{-s})\,\mathrm{d}s = \pi/2 + \log 2$ gives exactly the displayed D_Γ, c_Γ convention.

Here is the passage to $u=\kappa$, rather than a formal substitution. Every fixed derivative of κ decreases faster than every exponential. For a smooth physical function define

$$S_2(u) = \sum_{j=0}^{2}\int_{\mathbb{R}} e^{|x|}|u^{(j)}(x)|\,\mathrm{d}x.$$

Twice integrating by parts bounds $|\widehat{u}(z)|$ by a constant times $S_2(u)/(1+|z|)^2$ on $|\Im z| \le 1/2$. The unconditional zero-counting bound $N(T) = O(T\log(2+T))$ makes the polarized sum continuous under S_2 convergence of smooth cutoffs. The physical terms are continuous under H^1 convergence together with convergence of $e^{|x|}u$ in L^2: indeed

$$|\langle u, v(\cdot+s)\rangle| \le e^{-|s|}\|e^{|x|}u\|_2\|e^{|x|}v\|_2, \qquad \sum_{n\ge 2}\Lambda(n)n^{-3/2} < \infty.$$

The Gamma term is controlled by its small-shift derivative bound and its integrable large-shift tail. Theta cutoffs have all these convergences. Since Ξ vanishes at every z_{ρ_ζ} and its conjugate, (3.3) gives $\mathcal{Q}(\kappa, v) = 0$ for every compact smooth even v. The pole term is $2(\int C\kappa)(\int Cv) = \int Cv$, by (3.1). Thus the distributional identity is $D_\Gamma\kappa - c_\Gamma\kappa - P\kappa + C = 0$. The local continuity proved next makes it the pointwise identity (3.2). In particular, this argument does not infer radicality from a zero quadratic value in an indefinite form.

3.2 Derivative tails and a positive derivative source

On $x \ge 0$ put $t_j = \pi j^2 e^{2x}$ and $h_0(t) = 4t^2 - 6t$. Differentiation of a theta summand acts on its polynomial by

$$\mathcal{L}h = (\tfrac{1}{2} - 2t)h + 2th', \qquad h_k = \mathcal{L}^k h_0.$$

For any fixed k, h_k has degree $k+2$. A coefficient-sum bound, $t^q e^{-t} \le (2q)^q e^{-t/2}$, and $j^2 \ge 1+3(j-1)$ give constants A_k such that

$$|\kappa^{(k)}(x)| \le A_k e^{|x|/2}\frac{\exp[-(\pi/2)e^{2|x|}]}{1-e^{-3\pi/2}}. \tag{3.4}$$

Parity supplies negative x. This bounds the full series, not just its first term. Translated derivatives at $y \pm \log n$ consequently have Gaussian-in-n tails, uniformly on compact y intervals, even after

multiplication by $\Lambda(n)/\sqrt{n} \le \log n/\sqrt{n}$. The differentiated Gamma integrands are $O(s^2)$ near zero and are integrable at infinity. Thus the complete rows may be differentiated twice, and integration by parts against C has no boundary terms.

Define

$$r = \frac{\kappa'' + 20\kappa}{47}, \quad g_0 = \kappa - r = \frac{27\kappa - \kappa''}{47}, \quad m = \frac{81}{188}.$$

Lemma 3.1. *The following inequalities hold pointwise on the whole line:*

$$r \ge \kappa/188 > 0, \qquad r \ge \kappa \mathbf{1}_{\mathbb{R}\setminus I}, \qquad g_0 \le \kappa \mathbf{1}_I, \qquad r'' + 40r \ge 0.$$

Moreover,

$$\int_{\mathbb{R}} 2Cr \, \mathrm{d}x = m, \tag{3.5}$$

$$Pr = mC + (D_\Gamma - c_\Gamma)r, \tag{3.6}$$

$$Pg_0 = \frac{107}{188}C + (D_\Gamma - c_\Gamma)g_0. \tag{3.7}$$

Proof. Exact differentiation gives

$$h_2 = 16t^4 - 112t^3 + 165t^2 - \tfrac{75}{2}t,$$

$$h_4 = 64t^6 - 1056t^5 + 5176t^4 - 8512t^3 + \tfrac{15465}{4}t^2 - \tfrac{1875}{8}t.$$

The ratio

$$R_2(t) = h_2(t)/h_0(t) = 4t^2 - 22t + \tfrac{33}{4} + \frac{6}{2t-3}$$

satisfies $R_2'(t) \ge 2/3$ for $t \ge 3$, $R_2(3) = -79/4$, and $R_2(25/4) = 525/19$. All theta weights are positive, with $t_j > 3$. Hence $r/\kappa \ge (1/4)/47 = 1/188$. On $|x| \ge a$, $t_j \ge 2\pi > 25/4$, so $r/\kappa \ge (525/19 + 20)/47 = 905/893 > 1$.

For the last differential inequality,

$$h_4 + 60h_2 + 800h_0 = (t/8)Q(t),$$

$$Q(t) = 512t^5 - 8448t^4 + 49088t^3 - 121856t^2 + 135730t - 58275.$$

On the three intervals $[j, j+1]$, $j = 3, 4, 5$, write $5Q(j+u) = \sum_{i=0}^{5} B_{j,i}\binom{5}{i}u^i(1-u)^{5-i}$. The exact coefficient rows are

j	$B_{j,0}$	$B_{j,1}$	$B_{j,2}$	$B_{j,3}$	$B_{j,4}$	$B_{j,5}$
3	88575	113521	139459	163285	181127	190905
4	190905	200683	202397	193967	175105	149875
5	149875	124645	93047	59145	31355	25005

and are positive. For $u \ge 0$,

$$Q(6+u) = 512u^5 + 6912u^4 + 30656u^3 + 42880u^2 - 6350u + 5001.$$

The final quadratic is positive since $4 \cdot 42880 \cdot 5001 - 6350^2 = 817449020 > 0$. Thus $Q > 0$ on $[3, \infty)$, and summation proves $r'' + 40r \ge 0$.

Two integrations by parts give $\int 2C\kappa'' = 1/4$, so (3.5) follows from (3.1). Applying $(\partial_x^2 + 20)/47$ to (3.2), using $C'' = C/4$ and the convergence just proved, gives (3.6). Subtraction gives (3.7). □

3.3 A complete Gamma correction and weighted rows

Lemma 3.2. *If $v \ge 0$ is C^2, $v'' + 40v \ge 0$, and its compensated Gamma row exists, then*

$$(D_\Gamma - c_\Gamma)v \le -\frac{37}{350}v.$$

Consequently,

$$Pr \le mC - \frac{37}{350}r \le mC, \qquad P\kappa \le C - \frac{37}{350}\kappa < C, \qquad q_P := \frac{P\kappa}{2C} < \frac{1}{2}. \tag{3.8}$$

Also

$$\kappa(x)e^{|x|} \le 282r(x), \qquad B_L := P_-(\kappa e^{|\cdot|}) \le 282mC. \tag{3.9}$$

Proof. Put $k = \sqrt{40}$ and $F_x(s) = v(x+s) + v(x-s) - 2v(x)\cos(ks)$. Then $F_x(0) = F_x'(0) = 0$ and $F_x'' + 40F_x \ge 0$. Variation of constants, whose sine kernel is nonnegative for $0 \le s \le \pi/k$, gives $F_x \ge 0$ there. In particular, on $0 \le s \le 1/4$,

$$2v(x) - v(x+s) - v(x-s) \le 40s^2 v(x).$$

For larger s use $2v(x)$. Since $\rho(s) = e^{s/2}/(2\sinh s) < 4/(7s)$ on $(0, 1/4]$,

$$40\int_0^{1/4} s^2\rho(s)\,\mathrm{d}s < 5/7.$$

With $u = e^{-1/8} < 8/9$ the exact antiderivative gives

$$\int_{1/4}^{\infty} \rho(s)\,\mathrm{d}s = \operatorname{atanh} u + \arctan u,$$

so $D_\Gamma v \le (5/7 + \log 17 + \pi/2)v$. Now $\gamma > H_6 - \log 7 > 1/2$, because $H_6 = 49/20$ and $\log 7 < 39/20$. Also $8\pi > 25$ and $\log(25/17) > 8/25$ (integrate $1/t$ over $[17, 25]$). Therefore

$$c_\Gamma - (5/7 + \log 17 + \pi/2) > \tfrac{1}{2} + \tfrac{8}{25} - \tfrac{5}{7} = \frac{37}{350}.$$

This proves the comparison. It applies to r by Lemma 3.1, and to κ because $\kappa'' + 20\kappa \ge 0$ and $\kappa > 0$. Equations (3.2) and (3.6) prove (3.8).

For $|x| \le a$, $r \ge \kappa/188$ and $e^{|x|} \le \sqrt{2} < 3/2$. For $x \ge a$, the same positive theta weights give

$$\frac{r}{\kappa} \ge \frac{A(\pi e^{2x})}{47}, \qquad A(t) = R_2(t) + 20 = 4t^2 - 22t + \tfrac{113}{4} + \frac{6}{2t-3}.$$

For $t \ge 25/4$, $A(t) > t^2$: the polynomial part of $A(t) - t^2$ is increasing there and equals $127/16$ at $25/4$. Hence $r/\kappa > (9/47)e^{4x}$. Evenness supplies negative x, and the global comparison $\kappa e^{|x|} \le 282r$ follows. Positivity of the complete row gives $B_L \le 282Pr \le 282mC$. □

3.4 The core mass

Lemma 3.3. *For $a = \log 2/2$ and $I = [-a, a]$,*

$$K_I := \int_I \kappa(x)\,\mathrm{d}x > \frac{5780}{14391} > \frac{2}{5}.$$

Proof. For $h_1 = \mathcal{L}h_0 = -8t^3 + 30t^2 - 15t$,

$$h_1 + 7h_0 = -t(8t^2 - 58t + 57).$$

The quadratic is positive and increasing for $t \geq 25/4$; its value there is 7. Thus $\kappa' + 7\kappa < 0$ for $x \geq a$ and $\kappa(x) \leq \kappa(a)e^{-7(x-a)}$. At a, the first theta term is less than

$$\frac{6}{5}\frac{475}{4}e^{-25/4} < \frac{684}{2465},$$

since $e^{25/4} > 400(1 + 1/4 + 1/32 + 1/384) = 12325/24$. The complete remaining tail is bounded by

$$\frac{1536}{5}\sum_{j\geq 2} j^4 20^{-2j^2} < \frac{49152}{5 \cdot 20^8} < 10^{-6}.$$

Its successive ratio is below $1/2$. Since $5/18 - 684/2465 = 13/44370 > 10^{-6}$, $\kappa(a) < 5/18$. With $e^{a/2} < 6/5$ we obtain

$$\int_{\mathbb{R}\setminus I} 2C\kappa < \frac{5}{9}\left(\frac{6/5}{13/2} + \frac{1}{15/2}\right) = \frac{62}{351}.$$

Moreover $C(a)^2 = 1/2 + 3/(4\sqrt{2}) < 577/560 < (41/40)^2$, using $\sqrt{2} > 140/99$. Equation (3.1) yields

$$\frac{41}{20}K_I > \int_I 2C\kappa > \frac{289}{351},$$

which is the claimed bound. □

3.5 A uniform returning-core lower row

Lemma 3.4. *Set $q_I = (2C)^{-1}P(\kappa \mathbf{1}_I)$. Then*

$$q_I(y) > 1/16 \quad (y \geq a), \qquad L(y) = P_-\kappa(y) \geq C(y)/8 \quad (y \geq a).$$

In particular this lower bound for L is asserted only on the indicated half-line, not on all positive hubs.

Proof. First, the positive first theta term $F(x) = e^{x/2}h_0(\pi e^{2x})e^{-\pi e^{2x}}$, $x \geq 0$, has no interior minimum on $[0, b]$ for any b: the derivative polynomial $h_1(t) = t(-8t^2 + 30t - 15)$ has at most one sign change on $t \geq \pi$, from positive to negative. Endpoint bounds therefore give

$$\kappa(x) > 21/80 \ (|x| \leq a), \quad \kappa(x) > 1/3 \ (|x| \leq 3/10), \quad \kappa(x) > 2/5 \ (|x| \leq 13/50). \tag{3.10}$$

Here are explicit endpoint certificates. At zero, $F(0) > (325/16)/(63/2) = 325/504 > 2/5$. At a, $h_0(2\pi) > 475/4$, $2^{1/4} > 19/16$ and $e^{2\pi} < e^{44/7} < 537$, giving $F(a) > 21/80$. At $13/50$, $e^{13/25} < 17/10$ implies $\pi e^{13/25} < 27/5$. Since $h_0(t)e^{-t}$ decreases for $t \geq 3$,

$$F(13/50) > \frac{113}{100}\frac{2106/25}{225} > 2/5.$$

We used $e^{13/100} > 113/100$ and $e^{27/5} < 225$. Similarly, $e^{3/5} < 11/6$, $\pi e^{3/5} < 29/5$, $e^{3/20} > 23/20$ and $e^{29/5} < 333$ yield $F(3/10) > (23/20)(2494/25)/333 > 1/3$.

Let $a_2 = \log 2/\sqrt{2} > 49/100$. On $[a, 2/5]$, the $n = 2$ branch returns to I with displacement at most a, and $d(y) = 1/(2C(y)) > 49/100$. Thus

$$q_I > (49/100)^2(21/80) > 1/16.$$

On $[2/5, 9/20]$ its displacement is below $3/10$, while $d > 2/5$, so $q_I > (2/5)(49/100)(1/3) = 49/750 > 1/16$. The four closed intervals

$$[\log p - 13/50, \log p + 13/50], \qquad p = 2, 3, 5, 7,$$

cover $[9/20, 2]$. Indeed $\log 2 < 69315/100000$ places the first left endpoint below $9/20$; the largest adjacent prime ratio is $5/3$ and $e^{13/25} > 5/3$; and $\log 7 > 7/4$ places the last right endpoint above 2. Since $13/50 < a$, each chosen branch genuinely returns to I. On this interval $d > 8/25$ because $\cosh 1 < 25/16$. Also $\log p/\sqrt{p} \geq a_2$: the function $\log t/\sqrt{t}$ increases on $[2, 7]$, as $e^2 > 7$. Thus

$$q_I > (8/25)(49/100)(2/5) = 196/3125 > 1/16.$$

Additional primes and prime powers only increase these lower bounds.

It remains to cover every $y \geq 2$. The signed minorant g_0 from Lemma 3.1 obeys

$$|g_0(x)|,\ |g_0''(x)| < \varepsilon := 1/100 \qquad (|x| \geq 3/2). \tag{3.11}$$

For an explicit full-tail proof, at $t \geq 60$ coefficient-sum bounds give $|h_2| \leq 18t^4$, $|h_4| \leq 84t^6$, and

$$|27h_0 - h_2|/47 \leq 2t^6, \qquad |27h_2 - h_4|/47 \leq 2t^6.$$

For $x \geq 3/2$, $t_j > 60j^2$ and $e^{x/2} \leq t_j$. The bound $t^7 e^{-t} \leq 14^7 e^{-t/2}$ and the full geometric sum give both quantities in (3.11) an upper bound

$$\frac{2 \cdot 14^7 e^{-30}}{1 - e^{-90}} < \frac{2 \cdot 14^7}{20^{10}(1 - 20^{-30})} < 1/100.$$

Evenness treats $x < 0$.

For $y \geq 2$, keep the Gamma difference coupled on $0 < s \leq 1/2$. Taylor's integral remainder gives $2g_0(y) - g_0(y + s) - g_0(y - s) \geq -\varepsilon s^2$. Since $\rho(s) \leq 2/(3s)$ there, its integrated cost is $\varepsilon/12$. For $s \geq 1/2$, the inequalities $g_0 \leq \kappa \mathbf{1}_I$ and $g_0(y) \geq -\varepsilon$ give

$$2g_0(y) - g_0(y + s) - g_0(y - s) \geq -2\varepsilon - \kappa(y - s)\mathbf{1}_I(y - s).$$

Here $\int_{1/2}^{\infty} \rho < 4$; the returning-core term is at most $1/2$, because its shifts exceed $3/2$, where $\rho < 1$, and $\int_I \kappa \leq \frac{1}{2} \int 2C\kappa = 1/2$. The negative $2g_0(y)$ contribution has thus been retained, and

$$D_\Gamma g_0(y) \geq -\varepsilon/12 - 8\varepsilon - 1/2.$$

As $g_0(y) \leq 0$, $c_\Gamma > 0$, and $d(y) \leq 1/3$ on $y \geq 2$, positivity of the complete prime row and (3.7) yield

$$q_I \geq dPg_0 \geq \frac{107}{376} - \frac{1}{6} - \frac{97}{3600} = \frac{15391}{169200} > \frac{1}{16}.$$

Finally, for $y \geq a$ no $y + \log n$ lies in I. All returning-core branches therefore point left, and $L \geq P(\kappa \mathbf{1}_I) = 2Cq_I \geq C/8$. □

3.6 An outward weighted row and scalar certificates

The other weighted row needed in the hub estimate also has a complete analytic bound. By evenness, for $t \geq 0$,

$$\frac{B_L(-t)}{L(-t)} = e^t \frac{\sum_{n \geq 2} n a_n \kappa(t + \log n)}{R(t)} < 3e^t.$$

To check the last inequality, subtract the exact $n=2$ contribution $2R$ before bounding the remaining numerator. Put $T=\pi e^{2t}>3$ and $\phi(u)=(4u^2-6u)e^{-u}$. The positive $n=2, j=1$ denominator, $\phi(4T)\ge 56T^2e^{-4T}$, and $\phi(Tj^2n^2)\le 4T^2j^4n^4e^{-Tj^2n^2}$ give

$$\frac{\sum na_n\kappa(t+\log n)}{R(t)}\le 2+\frac{1}{7}\sum_{n\ge 3,j\ge 1} j^4n^6 20^{-(j^2n^2-4)}<3.$$

Indeed the j ratio is at most $16/20^{27}<1/2$ and the remaining n ratio is at most $(4/3)^6/20^7<1/2$; the double sum is at most $4\cdot 3^6/20^5=729/800000<1$. With $h_L=W(R/L)B_L$ this implies, using $dR<1/2$,

$$h_L(-t)<\tfrac{3}{2}e^t\kappa(t)\quad(t\ge 0),\qquad h_L(y)\le 1128m\kappa(y)=486\kappa(y)\quad(y\ge a).\tag{3.12}$$

For the second inequality combine $B_L\le 282mC$, $L\ge C/8$, $R<C$, and $W=\kappa/(2C)$. Reflection gives $h_R(y)=h_L(-y)$.

All auxiliary scalar brackets above can be checked by finite rational series, without evaluating a theta or prime sum. A convenient general certificate, for rational $x>0$ and $x<n+2$, is

$$E_n(x):=\sum_{j=0}^{n}\frac{x^j}{j!}<e^x\le U_n(x):=E_n(x)+\frac{x^{n+1}}{(n+1)!}\frac{1}{1-x/(n+2)}.$$

For the less familiar upper brackets used in this section, direct rational substitution gives

x	n	$U_n(x)$ is less than
44/7	50	537
13/25	12	17/10
27/5	40	225
3/5	12	11/6
29/5	40	333
1/3	12	3/2
1/4	12	4/3
1/8	12	8/7
2	25	8

The lower brackets $E_8(13/25)>5/3$, $E_6(39/20)>7$, $E_4(2)=7$, $E_3(3/4)>2$, and $E_4(3/2)>4$ are likewise rational. In particular $e^{7/4}=e^2/e^{1/4}<8/(5/4)=32/5<7$. The positive series

$$\log 2=2\sum_{j\ge 0}\frac{1}{(2j+1)3^{2j+1}}$$

with its geometric remainder gives $693/1000<\log 2<69315/100000$ after seven terms. Together with $\sqrt{2}<99/70$ it proves $a_2>49/100$. Finally, positive even Taylor sums and their geometric remainders give $\cosh(1/5)<50/49$, $\cosh(1/2)<8/7$, and $\cosh 1<25/16$. For example the first and last follow from

$$\sum_{j=0}^{4}\frac{(1/5)^{2j}}{(2j)!}+\frac{(1/5)^{10}}{10!}\frac{1}{1-1/(25\cdot 11\cdot 12)}<50/49$$

and

$$\sum_{j=0}^{6}\frac{1}{(2j)!}+\frac{1}{14!}\frac{1}{1-1/(15\cdot 16)}<25/16.$$

These certificates, the elementary $25/8<\pi<22/7$, and rational squaring establish all constants in this section independently of any source samples or numerical spectral premises.

4 The anchor and the minimum completion

Lemma 4.1. *For $g = f - \bar{f}_I$ and $u = \kappa g$,*

$$\mathcal{E}_\Gamma(f) \geq \int_{\mathbb{R}} A_{\mathrm{lo}}(x)|u(x)|^2\,\mathrm{d}x, \qquad A_{\mathrm{lo}}(x) = \frac{2\rho(s(x))}{5\kappa(x)}, \quad s(x) = \max\{\log 2, |x| + a\}. \tag{4.1}$$

Also, with $w(x) = \kappa(x)e^{-|x|/2}$,

$$\int_{\mathbb{R}} w|g|^2 \leq A_0\mathcal{E}_\Gamma(f), \qquad A_0 = \frac{e^a}{K_I} < \frac{15}{4}. \tag{4.2}$$

Proof. The function ρ is decreasing. The ordered edges within I contribute at least

$$\rho(2a)\left(K_I\int_I \kappa|g|^2 - \left|\int_I \kappa g\right|^2\right) = K_I\rho(2a)\int_I \kappa|g|^2.$$

For each exterior point x, the separate edges joining it to I contribute at least

$$\kappa(x)\rho(|x| + a)\left(K_I|g(x)|^2 + \int_I \kappa|g|^2\right).$$

The mixed term vanishes by the anchor. These two edge classes are disjoint and each edge is counted once. Dropping the extra nonnegative term yields $\mathcal{E}_\Gamma(f) \geq K_I\int \kappa\rho(s)|g|^2$. The mass bound $K_I > 2/5$ from Section 3 proves (4.1). Since $s(x) \leq |x| + 2a$ and $\rho(s) \geq e^{-s/2}$, it also proves (4.2). Finally $e^a = \sqrt{2} < 3/2$. □

If (f_j) is energy Cauchy, the anchored sequence $g_j = f_j - \bar{f}_{j,I}$ is Cauchy in $L^2(w)$ by (4.2). Since w is positive and continuous, its limit g is locally square integrable, even, and anchored. Choose a subsequence converging almost everywhere. Fatou's lemma on the nonnegative edge measure gives, for each j,

$$\mathcal{E}_\Gamma(g_j - g) \leq \liminf_k \mathcal{E}_\Gamma(g_j - g_k).$$

Thus $g_j \to g$ in energy and the completion norm equals its integral energy. A zero-energy representative is constant almost everywhere because the edge density is positive; the anchor makes that constant zero. This proves injectivity and does not assert that every anchored finite-energy measurable function lies in $H_\mathcal{E}$.

5 Five bounds for the retained map

Set $Y = 3/2$ and $N = 64$. The retained map is

$$T_c f(y) = \mathbf{1}_{\{|y|<Y\}}\sum_{2\leq n\leq N} a_n\left[\sqrt{\frac{WR}{L}}(y)u(y - \log n) - \sqrt{\frac{WL}{R}}(y)u(y + \log n)\right], \qquad u = \kappa(f - \bar{f}_I). \tag{5.1}$$

Centering occurs before numerator truncation. The finite uncentered numerator would not annihilate constants.

Write $v = A_{\mathrm{lo}}^{1/2}u$. Even input and odd output are identified isometrically with half-line functions by multiplying each restriction by $\sqrt{2}$. These factors cancel. On $0 < y < Y$ the folded comparison operator is therefore

$$Fv(y) = \sum_{n\leq 64}\{q_{-,n}(y)v(|y - \log n|) - q_{+,n}(y)v(y + \log n)\}, \tag{5.2}$$

where the nonnegative coefficients satisfy exactly

$$p_{-,n} = \frac{a_n \kappa(y - \log n)}{L(y)}, \qquad p_{+,n} = \frac{a_n \kappa(y + \log n)}{R(y)}, \tag{5.3}$$

$$q_{-,n}^2 = \frac{5}{2} \frac{a_n W R\, p_{-,n}}{\rho(s(y - \log n))}, \qquad q_{+,n}^2 = \frac{5}{2} \frac{a_n W L\, p_{+,n}}{\rho(s(y + \log n))}. \tag{5.4}$$

The retained sums of each family of p's are at most one. A one-to-one translation or reflection has norm bounded by the supremum of its coefficient. The high-left part is not one-to-one; both possible preimages will be counted explicitly.

Proposition 5.1. *As operators on the half-line input space, the five pieces of* (5.2) *have the following norm bounds:*

Piece	*Strict upper bound*
Right branch $n = 2$	$3/10$
All right branches $n \geq 3$	$1/1000$
Left branch $n = 2$ *on* $0 < y < a$	$1/25$
All left branches $n \geq 3$ *on* $0 < y < a$	$1/1000$
Retained left branches $n \leq 64$ *on* $a < y < 3/2$	$7/1000$

The right and low-left bounds may include every label, not just those through 64*.*

Proof. The right branch $n = 2$. Here $s(y + \log n) = y + \log n + a$ and $1/\rho(s) \leq e^{s/2}$. Hence

$$\begin{aligned} q_{+,2}^2 &\leq F_0(y) := \tfrac{5}{2}(\log 2) 2^{1/4} e^{y/2} W(y) L(y) \\ &\leq \tfrac{5}{4}(\log 2) 2^{1/4} \kappa(y) e^{y/2} L(y). \end{aligned} \tag{5.5}$$

For every $n \geq 2$ and $y \geq 0$,

$$\kappa(y)\kappa(y - \log n) e^{y/2} \leq K^2 n^{19/4} \exp\left(-2\pi n + \frac{1}{64\pi n}\right). \tag{5.6}$$

Indeed, for $0 \leq y \leq \log n$ put $t = 2y - \log n$. The envelope product is $K^2 n^{19/4} e^{t/4 - 2\pi n \cosh t}$, and

$$t/4 - 2\pi n \cosh t \leq -2\pi n - \pi n t^2 + t/4 \leq -2\pi n + 1/(64\pi n).$$

For $y \geq \log n$ its logarithmic derivative is $19/2 - 2\pi(1 + n^{-2}) e^{2y} < 0$, so the endpoint already treated controls it. Using $2^{1/4} < 6/5$, $K < 40$, and $e^{1/(64\pi n)} < 400/399$, (5.5) gives

$$\sup F_0 \leq 1680 \frac{400}{399} \sum_{n \geq 2} \Lambda(n) n^{17/4} e^{-2\pi n}.$$

The $n = 2$ term is less than

$$H_2 = \frac{(7/10)(96/5)}{20^4 (79/48)}.$$

For $n \geq 3$, use $\Lambda(n) \leq n/2$, $3^{1/4} < 4/3$, and a successive ratio at most $(4/3)^6/500 < 3/250$. Since $e^{6\pi} > 20^6 (269/128)$, the rest is less than

$$H_* = \frac{162}{20^6 (269/128)} \frac{250}{247}.$$

Consequently

$$\sup F_0 < 1680\frac{400}{399}(H_2 + H_*) = \frac{1097188704}{12466367875} < \frac{9}{100}. \tag{5.7}$$

The one-to-one translation gives norm less than 3/10.

The remaining right branches. Equation (5.4) gives $q_{+,n}^2 \le (\Lambda(n)/\log 2)F_0 p_{+,n}$. Combining (2.3) and (5.7), and using $\sqrt{8/7} < 8/7$ and $\Lambda(n)/\log 2 < n$, yields

$$\sup_y q_{+,n}(y) \le \frac{3}{35} n^3 e^{-(\pi/2)(n^2-4)} \quad (n \ge 3).$$

The leading exponential denominator exceeds 2400, since $e^{5\pi/2} > e^6 e^{29/16} > 20^2 \cdot 6$. The successive ratio is at most $(4/3)^3/20^3 = 1/3375 < 1/1000$. The triangle inequality therefore gives

$$\sum_{n\ge 3} \sup q_{+,n} < \frac{3}{35}\frac{27}{2400}\frac{1000}{999} = \frac{1}{1036} < \frac{1}{1000}. \tag{5.8}$$

The low-left branch $n = 2$. On $0 < y < a$, $r_n = \log n - y \ge a$ and $s(r_n) = r_n + a$. Using $p_{-,2} \le 1$, $W(y)e^{-y/2} = \kappa(y)/(1 + e^y)$, $\kappa < 1$, and $R(y) \le R(0)$, we get

$$q_{-,2}^2 \le \frac{5}{4}(\log 2)2^{1/4}R(0) < \frac{5}{4}\frac{7}{10}\frac{6}{5}\frac{3}{2000} = \frac{63}{40000} < \frac{1}{625}.$$

The reflection is one-to-one, giving the bound 1/25.

The remaining low-left branches. Put $r_2 = \log 2 - y$, $r_+ = \log 2 + y$. Since $r_2 \ge a$, the envelope and the first denominator summand imply

$$\frac{\kappa(\log n - y)}{\kappa(r_2)} \le \frac{4}{3}\left(\frac{n}{2}\right)^{9/2} e^{-\pi e^{-2y}(n^2-4)}. \tag{5.9}$$

Here $(500/499)/(1 - 3/(4\pi)) < (500/499)(25/19) < 4/3$. Retain $L \ge a_2\kappa(r_2)$, use $R < (1001/1000)a_2\kappa(r_+)$, and keep the shared y dependence before taking any supremum. With $v = e^{2y} \in [1, 2]$, (5.4) gives

$$q_{-,n}^2 \le \frac{5}{4}\frac{1001}{1000}\frac{4}{3}K2^{1/4}\Lambda(n)^2 n^4 v^2 \exp\left[-\pi\left(4v + \frac{n^2-4}{v}\right)\right]. \tag{5.10}$$

The factors use $W \le 1/2$; the powers $2^{9/2}$ cancel, and the remaining exponential prefactor is $e^{4y} = v^2$. Since $v^2 \le 4$, $4v + (n^2-4)/v \ge 4\sqrt{n^2-4}$, and $5(1001/1000)(4/3)40(6/5) = 8008/25 < 18^2$,

$$\sup_{0<y<a} q_{-,n}(y) < 18\Lambda(n)n^2 e^{-2\pi\sqrt{n^2-4}} \le 9n^3 e^{-2\pi(n-1)} \quad (n \ge 3).$$

The last inequality uses $\sqrt{n^2-4} \ge n - 1$ and $\Lambda(n) \le n/2$. The leading term and a geometric ratio give

$$\sum_{n\ge 3} \sup_{0<y<a} q_{-,n} < \frac{243}{20^4(79/48)}\frac{1}{1 - 64/13500} = \frac{19683}{21228880} < \frac{1}{1000}. \tag{5.11}$$

The high-left branches and their folds. Let H be the retained left operator on $a < y < Y$. Conditional Cauchy–Schwarz with $\sum_{n\le 64} p_{-,n} \le 1$ yields

$$|Hv(y)|^2 \le \sum_{n\le 64}\frac{5}{2}\frac{a_n W(y)R(y)}{\rho(s(y - \log n))}|v(|y - \log n|)|^2.$$

For a fixed positive input point x a label has at most two preimages, $y = \log n + x$ and $y = \log n - x$, each with Jacobian magnitude one. The bounds

$$\rho(s(x))^{-1} \le \sqrt{2}e^{x/2}, \quad x \le y + \log n, \quad W(y)e^{y/2} = \frac{\kappa(y)}{1 + e^{-y}} < 1$$

show that each preimage costs at most $(5/2)\sqrt{2}\Lambda(n)R(a) < (15/4)\Lambda(n)R(a)$. Counting both and using the elementary Chebyshev bound $\sum_{n\le t} \Lambda(n) \le 4(\log 2)t$, proved in Lemma 6.1, gives

$$\|H\|^2 \le \frac{15}{2} R(a) \sum_{n\le 64} \Lambda(n) < \frac{15}{2} \frac{1}{30000000} \frac{896}{5} = \frac{7}{156250} < \frac{49}{10^6}.$$

This is the only one of the five estimates that requires the finite label cutoff. □

The low-left and high-left outputs have disjoint supports. Their combined norm is therefore less than

$$\sqrt{(41/1000)^2 + (7/1000)^2} < 42/1000,$$

since $41^2 + 7^2 = 1730 < 1764 = 42^2$. The combined right norm is less than $301/1000$. Together with (4.1) and the two half-line isometries, this proves

$$\|T_c f\|_2 \le \frac{343}{1000} \mathcal{E}_\Gamma(f)^{1/2}. \tag{5.12}$$

The comparison is on a larger weighted input space only to obtain an upper bound. The actual input remains the anchored representative in $H_\mathcal{E}$.

6 Restoring every output point and every label

Put

$$B_L = P_-(\kappa e^{|\cdot|}), \quad B_R = P_+(\kappa e^{|\cdot|}), \quad h_L = W\frac{R}{L}B_L, \quad h_R = W\frac{L}{R}B_R = h_L(-\cdot). \tag{6.1}$$

For any nonnegative output cutoff $\chi \le 1$, weighted Cauchy–Schwarz and Tonelli give

$$\|\chi^{1/2} T g\|_2^2 \le 2 \int_\mathbb{R} \kappa(x) e^{-|x|} |g(x)|^2 [P_+(\chi h_L) + P_-(\chi h_R)](x)\, \mathrm{d}x. \tag{6.2}$$

All sums in this step are positive. The inequality first holds for finite partial sums and then for their increasing absolute-sum majorants.

The source row estimates proved in Section 3 imply, for $r \ge a$,

$$h_L(r) \le 486\kappa(r), \qquad h_L(-r) < \tfrac{3}{2} e^r \kappa(r). \tag{6.3}$$

In particular, the inward estimate uses $L(r) \ge C(r)/8$ only in its valid domain $r \ge a$. Define, for $Y \ge a$,

$$v_Y(z) = \mathbf{1}_{\{|z|>Y\}} V(|z|), \qquad V(r) = 486 K e^{11r/2 - \pi e^{2r}}. \tag{6.4}$$

Both restricted functions $\mathbf{1}_{|y|>Y} h_L$ and $\mathbf{1}_{|y|>Y} h_R$ are bounded by v_Y. Its values at $\pm Y$ are zero.

6.1 A prime-row estimate with cutoff atoms

Lemma 6.1. *For $t \geq 1$,*

$$\psi(t) := \sum_{n \leq t} \Lambda(n) \leq Bt, \qquad B = 4 \log 2. \tag{6.5}$$

If $F \geq 0$ is piecewise C^1 with finitely many jumps, tends to zero at infinity, and has finite weighted total variation on $[1, \infty)$, let $\mu_\downarrow = (\mathrm{d}F)_-$ include its downward jump atoms. If each assigned jump value is at most the larger one-sided limit, then

$$\sum_{n \geq 2} \Lambda(n) F(n) \leq B \int_{[1,\infty)} t \, \mathrm{d}\mu_\downarrow(t). \tag{6.6}$$

Proof. For an integer $m \geq 1$, every prime power in $(m, 2m]$ contributes to the corresponding valuation of $\binom{2m}{m}$. Thus $\psi(2m) - \psi(m) \leq \log \binom{2m}{m} \leq 2m \log 2$. Sum at powers of two and round t upward to the next power of two to get (6.5). For the second assertion, integrating the signed variation to infinity bounds both one-sided values of $F(t)$ by $\mu_\downarrow([t, \infty))$; this also bounds any allowed assigned value. Tonelli gives

$$\sum_{n \geq 2} \Lambda(n) F(n) \leq \int_{[1,\infty)} \psi(t) \, \mathrm{d}\mu_\downarrow(t).$$

Use (6.5). The closed interval ensures that a prime power exactly at a downward jump is counted. There is no boundary contribution at 1, since $\Lambda(1) = 0$. □

For the functions below define

$$J_-(v) = \int_{\mathbb{R}} e^{-z/2} (\, \mathrm{d}v + (v/2) \, \mathrm{d}z)_+, \qquad J_+(v) = \int_{\mathbb{R}} e^{z/2} (\, \mathrm{d}v - (v/2) \, \mathrm{d}z)_-.$$

Apply (6.6) to $F_\mp(t) = t^{-1/2} v(x \mp \log t)$. Their derivatives away from jumps are respectively $-t^{-3/2}[v'(x - \log t) + v(x - \log t)/2]$ and $t^{-3/2}[v'(x + \log t) - v(x + \log t)/2]$. The same orientation reversal applies to the jump measures. Changing variables and enlarging a nonnegative integral gives

$$P_- v(x) \leq B e^{x/2} J_-(v), \qquad P_+ v(x) \leq B e^{-x/2} J_+(v). \tag{6.7}$$

For even v the two J's agree. In particular, $P v_Y \leq 2B J_\downarrow(Y) C$.

Since $V'(r) = (11/2 - 2\pi e^{2r}) V(r)$, the measure $\mathrm{d}v_Y + (v_Y/2) \, \mathrm{d}z$ has a negative atom $-V(Y)$ at $-Y$ and a positive atom $V(Y)$ at Y. Its density at $z = -r < -Y$ is $(2\pi e^{2r} - 5) V(r) > 0$, while its density at $z = r > Y$ is $(6 - 2\pi e^{2r}) V(r) < 0$. Hence integration by parts gives exactly

$$\begin{aligned} J_\downarrow(Y) &= e^{-Y/2} V(Y) + \int_Y^\infty e^{r/2} [-V'(r) + V(r)/2] \, \mathrm{d}r \\ &= V(Y)(e^{Y/2} + e^{-Y/2}) + \int_Y^\infty e^{r/2} V(r) \, \mathrm{d}r \\ &\leq V(Y) \left(e^{Y/2} + e^{-Y/2} + \frac{e^{Y/2}}{2\pi e^{2Y} - 6} \right). \end{aligned} \tag{6.8}$$

The last step uses the logarithmic derivative $6 - 2\pi e^{2r} \leq -(2\pi e^{2Y} - 6) < 0$ of $e^{r/2} V(r)$. The derivative signs and jump atoms are both required for this estimate.

6.2 The output tail

Use (6.2), (6.7), and $e^{-|x|}C(x) \le e^{-|x|/2}$, then (4.2), to obtain

$$\|\mathbf{1}_{|y|>Y}Tg\|_2^2 \le C_{\mathrm{hub}}(Y)\mathcal{E}_\Gamma(g), \qquad C_{\mathrm{hub}}(Y) = 4BA_0V(Y)\left(e^{Y/2}+e^{-Y/2}+\frac{e^{Y/2}}{2\pi e^{2Y}-6}\right). \tag{6.9}$$

This is $O(e^{6Y-\pi e^{2Y}})$ as $Y\to\infty$. At $Y=3/2$, $486K<19440$, $4BA_0<42$, and

$$\frac{33}{4}-\pi e^3 < \frac{33}{4}-\frac{25}{8}\cdot 20 = -\frac{217}{4} < -54.$$

Also $e^{3/4}<9/4$, $e^{-3/4}<1/2$, and $2\pi e^3-6>119$. Thus

$$C_{\mathrm{hub}}(3/2) < 42\frac{659}{238}\frac{19440}{20^{18}} = \frac{480411}{5570560000000000000000} < 9\cdot 10^{-18}. \tag{6.10}$$

6.3 The label tail with complete denominators

For general $Y\ge a$ and an integer $N\ge e^{2Y}$, let $T_{Y,N}$ be defined by (5.1) with these cutoffs. The omitted label rows satisfy, for $|y|\le Y$,

$$B_{\pm,>N}(y) := \sum_{n>N} a_n\kappa(y\pm\log n)e^{|y\pm\log n|} \le \beta(Y,N),$$
$$\beta(Y,N) = 256e^{11Y/2}(N+1)^6e^{-3e^{-2Y}(N+1)^2}. \tag{6.11}$$

Indeed (2.2) implies $\kappa(s)e^s \le 128e^{11s/2-3e^{2s}}$ for $s\ge 0$. Using $\Lambda(n)\le n$ bounds the series by $128e^{11Y/2}\sum_{n>N} n^6e^{-3e^{-2Y}n^2}$. Its successive ratio is at most $64e^{-3e^{-2Y}(2N+3)} \le 64e^{-6} < 1/2$, proving (6.11).

Keeping only $n=2, j=1$ gives a uniform positive lower bound for both complete denominators:

$$L(y), R(y) \ge \ell_Y := \frac{441}{50}e^{-16e^{2Y}} \qquad (|y|\le Y). \tag{6.12}$$

For $|x|\le D$ one has $\kappa(x)\ge 18e^{-4e^{2D}}$ by the first theta term. Set $D=Y+\log 2$ and use $a_2>49/100$. The last comparison follows, for example, from $\log 2 > 2(1/3+1/81+1/1215) > 693/1000$ and $\sqrt{2}<99/70$.

By the source estimates $r\ge\kappa/188$, $Pr\le mC$ with $m=81/188$, and $L,R\le C$, we have $W/r\le 94/C$. Therefore

$$\mathbf{1}_{|y|\le Y}W\frac{R}{L}B_{-,>N} \le \epsilon(Y,N)r, \quad \mathbf{1}_{|y|\le Y}W\frac{L}{R}B_{+,>N}\le\epsilon(Y,N)r, \quad \epsilon=\frac{94\beta}{\ell_Y}.$$

Weighted Cauchy–Schwarz on the omitted labels, followed by positive transposition, bounds the remaining transposed row by Pr. It follows that

$$\|\mathbf{1}_{|y|\le Y}T - T_{Y,N}\|^2_{H_{\mathcal{E}}\to L^2} \le 2mA_0\epsilon(Y,N)$$
$$< \frac{432000}{49}e^{11Y/2}(N+1)^6e^{16e^{2Y}-3e^{-2Y}(N+1)^2}. \tag{6.13}$$

This is an all-label estimate, and neither denominator has been replaced by a finite row.

At $Y=3/2, N=64$, admissibility follows from $e^3<21<64$. Since

$$16e^3-12675e^{-3} < 336-\frac{12675}{21} = -\frac{1873}{7} < -267,$$

and $e^{33/4}<e^9<3^9$, $e^{267}>20^{89}$, we obtain

$$\|\mathbf{1}_{|y|\le 3/2}T-T_c\|^2 < \frac{432000}{49}\frac{3^9\,65^6}{20^{89}} < 10^{-96}. \tag{6.14}$$

7 Completion of the proof and scope

On the smooth core, combine (5.12) and (6.14) on $|y| \le 3/2$. The output tail is orthogonal to that entire part, so (6.10) gives

$$\|Tf\|_2^2 \le \left[\left(\frac{343}{1000} + 10^{-48}\right)^2 + 9 \cdot 10^{-18}\right] \mathcal{E}_\Gamma(f).$$

The last coefficient is strictly less than $3/25 < 1/8$ by an exact rational comparison. The core and label errors share their output support, so their norms must be added before squaring; omitting their cross term would be incorrect.

These bounds extend the core map continuously to the minimum completion. We also verify that the extension has the asserted full-row interpretation. The functions h_L, h_R are continuous and finite on compact intervals: positive complete source tails converge locally uniformly, and the denominators are positive. On the exterior, (6.3) and the positive lower bound for r in Section 3 show that h_L/r and h_R/r are bounded. On compact intervals their ratios are bounded by positivity of r. Thus there is a finite D with $h_L, h_R \le Dr$ globally. Weighted Cauchy–Schwarz for absolute partial sums gives

$$\int_{\mathbb{R}} \left[W\frac{R}{L}|P_-(\kappa g)|^2 + W\frac{L}{R}|P_+(\kappa g)|^2\right] \le D\int_{\mathbb{R}} \kappa e^{-|x|}|g|^2 Pr \le DmA_0\mathcal{E}_\Gamma(g),$$

with the same argument first applied to the absolute sums. It proves almost-everywhere absolute convergence of each first moment. Countably many translations preserve null sets. The resulting integral operator agrees with the continuous extension by density. Zero energy forces the anchored representative to vanish, as already shown in Section 4. This completes the proof of Theorem 1.1.

The role of 64 is limited to a numerator decomposition and one finite-label fold estimate. The theorem itself includes every label and every output point. No finite matrix, sampled norm, trial-space eigenvalue, or quadrature value is used. The estimates are not asserted to be sharp. The only strict final inequalities concern positive-energy inputs; the zero-energy case is stated separately. None of these conclusions addresses the sign of the remaining full signed operator.

A Elementary constants and reproducibility

For the upper bound on π, polynomial division gives

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,\mathrm{d}x = \frac{22}{7} - \pi > 0.$$

For the lower bound use $\arctan 1 = \arctan(1/2) + \arctan(1/3)$ and the finite geometric lower bounds for $1/(1+t^2)$:

$$\pi > 4\left[\frac{1}{2} - \frac{(1/2)^3}{3} + \frac{(1/2)^5}{5} - \frac{(1/2)^7}{7} + \frac{1}{3} - \frac{(1/3)^3}{3}\right] = \frac{284663}{90720} > \frac{25}{8}.$$

Positive Taylor partial sums give

$$e^3 > \sum_{k=0}^{8} \frac{3^k}{k!} = \frac{89641}{4480} > 20, \qquad e^{7/10} > \sum_{k=0}^{3} \frac{(7/10)^k}{k!} = \frac{12013}{6000} > 2.$$

The tail from the fifth term of e has ratio at most $1/6$, hence

$$e < \frac{65}{24} + \frac{1/120}{1-1/6} = \frac{1631}{600} < \frac{68}{25} < 3, \qquad (68/25)^3 < 21.$$

In particular $e^{1/2} < \sqrt{3} < 2$, so $\log 2 > 1/2$. Further positive partial sums give $e > 8/3$, $e^{1/2} > 79/48$, $e^{3/4} > 269/128$, and $e^{29/16} > \sum_{k=0}^{6} (29/16)^k/k! > 6$. The comparisons of square and fourth roots used in the text follow by taking the corresponding positive integer powers. For $0 < t < 1$, $e^t < 1/(1-t)$; this proves $e^{1/(64\pi n)} < 400/399$ for $n \geq 2$.

The accompanying Python sources use integer arithmetic and `fractions.Fraction` to check the displayed scalar margins. They do not evaluate κ, enumerate primes, compute an integral, approximate an operator norm, or verify the analytic inequalities between functions. Their successful execution is a reproducibility check for the rational arithmetic only. The mathematical proof also requires the source identities, complete infinite-series comparisons, edge argument, changes of variables, and completion argument given in the text.

Data availability

The PDF, editable LaTeX sources, and source-free exact-rational checks accompany this preprint. The repository is `https://github.com/serabing-hash/theta-weighted-certificate`; the directory for this version is `research/2026-10-27-directional-prime-mean-bound/`. No external numerical dataset is required.

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Author disclosure and manuscript status

Artificial intelligence assistance was used to organize the source arguments, draft and check the mathematical exposition, and prepare the LaTeX manuscript. Separate AI-assisted reviews of the source arguments are internal mathematical checks, not human peer review. This preprint makes no claim of journal acceptance, human peer review, or priority. The work reported here uses no new numerical theta or prime source evaluation, quadrature, spectral computation, or interval-source certificate. Its claim is the directional mean inequality on the specified anchored minimum completion, not a conclusion about the full signed operator or the Riemann hypothesis.

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